

The Rosenthal–Odell Quotient Problem

Audit of a Partial Proof, Literature Status, and Sharpened Reductions

22 June 2026

Abstract

Gasparis records two closely related questions: whether every quotient of an ℓ_1 -predual is c_0 -saturated, and, in Odell’s special case, whether every quotient of $C(K)$ is c_0 -saturated when K is countable compact. The phrase “ ℓ_1 -predual” has two conventions in the literature. Under the strict convention, meaning that the dual is *isometric* to ℓ_1 , I found no published resolution through 22 June 2026. Under the broader convention, meaning only that the dual is *isomorphic* to ℓ_1 , Argyros–Motakis provide a counterexample.

The supplied complemented-subspace theorem is correct. It is, however, subsumed by a stronger 2017 theorem of Galego–González–Pello: every quotient X of a real strict ℓ_1 -predual satisfies

$$\mathcal{U}(X, Z) = \mathcal{K}(X, Z) \quad \text{for every Banach space } Z,$$

where $\mathcal{U}$ and $\mathcal{K}$ denote the unconditionally converging and compact operators. This yields several additional consequences: every infinite-dimensional quotient of X contains a complemented copy of c_0 ; X has no infinite-dimensional reflexive quotient; and X is superprojective. The remaining saturation problem is reduced exactly to the opposite projection property:

$$X \text{ is } c_0\text{-saturated} \iff X \text{ is subprojective.}$$

The report also proves an iterative c_0 -splitting reduction, an extension obstruction based on the Josefson–Nissenzweig theorem, and a general (WU) -trichotomy. These sharply constrain a possible counterexample but do not close the strict problem.

1 The question and the convention issue

A Banach space is c_0 -saturated if every closed infinite-dimensional subspace contains an isomorphic copy of c_0 . Gasparis asks whether every quotient of an ℓ_1 -predual is c_0 -saturated and attributes to Odell the special case of quotients of $C(K)$ for countable compact metric K [6, p. 492]. The same problem appears in Alspach’s work, Rosenthal’s survey, and Argyros–Kanellopoulos [1, 14, 2].

The word “predual” is not used uniformly:

Definition 1.1. A *strict ℓ_1 -predual* is a Banach space E such that E^* is isometric to ℓ_1 (or to $\ell_1(\Gamma)$ in the nonseparable notation). A *broad, or isomorphic, ℓ_1 -predual* is a Banach space E such that E^* is merely isomorphic to ℓ_1 .

The strict interpretation is the one relevant to the source problem. One clear indication is the standard assertion used there that ℓ_1 -preduals are c_0 -saturated, which is an isometric L_1 -predual phenomenon. Alspach also explicitly adopts the isometric convention in his paper on quotients of ordinal C -spaces

[1]. By contrast, Argyros–Motakis explicitly define an ℓ_1 -predual to mean a space whose dual is only isomorphic to ℓ_1 [3, p. 36].

Throughout the strict part of this report, scalars are real. This is the setting in which the Johnson–Zippin property-(V) theorem is invoked in the supplied proof and in the cited quantitative formulation [9, Theorem 4.2].

2 Audit of the supplied partial proof

The supplied theorem is the following.

Theorem 2.1 (Supplied partial result). *Let E be a real Banach space such that E^* is isometric to ℓ_1 , and let X be a quotient of E . Every complemented infinite-dimensional subspace of X contains an isomorphic copy of c_0 .*

Consequently, the same conclusion holds when X is a quotient of $C(K)$ and K is countable compact metric.

Verdict

The proof is correct. Its logical chain is:

1. strict real ℓ_1 -preduals have Pełczyński’s property (V);
2. property (V) passes to quotients;
3. an operator is unconditionally converging exactly when it fixes no copy of c_0 ;
4. a projection onto a c_0 -free range is therefore weakly compact;
5. the identity on that range is weakly compact, so the range is reflexive;
6. the dual of the range embeds into $E^* = \ell_1$, hence has the Schur property; a reflexive Schur space is finite-dimensional.

There is one bibliographic correction: the relevant result in Krulišová’s paper is Theorem 4.2, not Theorem 4.5 [9, Theorem 4.2]. There is also a shorter final step, given in theorem 3.2: the projection is not only weakly compact but compact.

For completeness, the quotient inheritance used in the proof is sound. If $Q : E \twoheadrightarrow X$ and $T : X \rightarrow Z$ is unconditionally converging, then TQ is unconditionally converging. If TQ is weakly compact, the open mapping theorem gives a constant C with $B_X \subset Q(CB_E)$, and hence $T(B_X) \subset TQ(CB_E)$ is relatively weakly compact.

3 The stronger known operator theorem

The main improvement comes from combining property (V) with the Schur property of the dual. Galego–González–Pello package this as the class $\text{Sp}(\mathcal{U}^{-1} \circ \mathcal{K})$ and prove both that strict ℓ_1 -preduals belong to it and that the class is stable under quotients [5, Corollary 3.5 and Proposition 3.6].

Theorem 3.1 (Compactness of unconditionally converging operators). *Let E be a real Banach space with E^* isometric to ℓ_1 , and let $Q : E \rightarrow X$. Then, for every Banach space Z ,*

$$\mathcal{U}(X, Z) = \mathcal{K}(X, Z).$$

Equivalently, every noncompact operator $T : X \rightarrow Z$ fixes an isomorphic copy of c_0 .

Proof. The quotient X has property (V). Moreover, $Q^* : X^* \rightarrow E^*$ is an isomorphic embedding, so X^* is a subspace of a Schur space and therefore has the Schur property.

Let $T : X \rightarrow Z$ be unconditionally converging. Property (V) makes T weakly compact. By Gantmacher's theorem, $T^* : Z^* \rightarrow X^*$ is weakly compact. Relatively weakly compact subsets of a Schur space are relatively norm compact, so T^* is compact. Schauder's theorem then implies that T is compact. The reverse inclusion $\mathcal{K}(X, Z) \subset \mathcal{U}(X, Z)$ is standard. $\square$

This direct proof also explains why the word *isometric* matters: the Schur property survives in a subspace of ℓ_1 , but property (V) can fail for a broad ℓ_1 -predual. Galego–González–Pello explicitly note that “dual isometric” cannot be replaced by “dual isomorphic” in their corollary [5, Corollary 3.5].

Corollary 3.2 (Complemented subspaces). *Under the hypotheses of theorem 3.1, every complemented infinite-dimensional subspace of X contains c_0 . In fact, if $P : X \rightarrow X$ is a projection whose range contains no c_0 , then P has finite rank.*

Proof. If $P(X)$ contains no c_0 , then P fixes no copy of c_0 , so P is unconditionally converging. By theorem 3.1, P is compact. Since P is the identity on $P(X)$, the unit ball of $P(X)$ is norm compact, and $P(X)$ is finite-dimensional. $\square$

Thus the supplied argument is not merely right; its conclusion is a special case of a stronger operator ideal statement already in the literature.

Corollary 3.3 (Infinite-dimensional quotients). *Let $R : X \rightarrow W$ be a quotient map with W infinite-dimensional. Then R fixes a copy $A \simeq c_0$. The copy $R(A) \simeq c_0$ is complemented in W , and A may be chosen complemented in X . Consequently, X has no infinite-dimensional reflexive quotient.*

Proof. A compact surjection has finite-dimensional range, so R is noncompact. By theorem 3.1, R is not unconditionally converging and therefore fixes a copy $A \simeq c_0$. The spaces E , X , and W are separable because E^* is separable. Sobczyk's theorem complements $R(A)$ in W . If $P : W \rightarrow R(A)$ is a projection and $S : R(A) \rightarrow A$ is the inverse of $R|_A$, then $SPR : X \rightarrow A$ is a projection onto A .

If W were infinite-dimensional reflexive, it could not contain c_0 , contradicting the first part. $\square$

Corollary 3.4 (Superprojectivity). *Every quotient X of a strict real ℓ_1 -predual is superprojective.*

Proof. This is Corollary 3.4 together with Corollary 3.5 and Proposition 3.6 of [5]. It also follows directly from theorem 3.3: for an infinite-codimensional subspace $M \subset X$, the quotient map $X \rightarrow X/M$ fixes a copy of c_0 whose image is complemented in the separable quotient. Pulling back a complementary summand gives an infinite-codimensional complemented subspace of X containing M . $\square$

4 An exact reduction: saturation versus subprojectivity

Recall that X is *subprojective* if every infinite-dimensional subspace of X contains an infinite-dimensional subspace complemented in X .

Theorem 4.1 (Exact reduction). *Let X be a quotient of a real strict ℓ_1 -predual. Then*

$$X \text{ is } c_0\text{-saturated} \iff X \text{ is subprojective.}$$

The same equivalence holds for every quotient of $C(K)$ with K countable compact metric.

Proof. Suppose first that X is c_0 -saturated. If $Y \subset X$ is infinite-dimensional, choose $C \subset Y$ with $C \simeq c_0$. Since X is separable, Sobczyk's theorem makes C complemented in X . Thus X is subprojective. This implication is also a special case of a theorem of Díaz–Fernández, as recorded in [5, Proposition 2.7].

Conversely, suppose X is subprojective and let $Y \subset X$ be infinite-dimensional. There is an infinite-dimensional $Z \subset Y$ complemented in X . By theorem 3.2, Z , and hence Y , contains c_0 . $\square$

Accordingly, the strict Rosenthal–Odell problem can be restated as follows.

Equivalent strict problem. Is every quotient of a strict ℓ_1 -predual subprojective? In particular, is every quotient of $C(K)$ subprojective when K is countable compact?

This reformulation is useful because the *known* direction is the dual one: such quotients are superprojective by theorem 3.4. The 2025 work of González–Pello emphasizes that subprojectivity passes to subspaces, superprojectivity passes to quotients, and neither property is a general three-space property [7]. Thus the missing implication is not a formal permanence fact.

5 Literature search: what is solved and what remains

5.1 The strict/isometric version

I found no published proof or counterexample for the strict version through 22 June 2026. The following checkpoints support that assessment.

- Alspach wrote in 1992 that it was unknown whether quotients of $C(\alpha)$ must be c_0 -saturated [1].
- Odell proved that quotients of Schreier's space are c_0 -saturated, and listed the higher Schreier and countable- $C(K)$ questions as open [11].
- Rosenthal's 2003 survey states that the countable- $C(K)$ quotient problem is unknown and notes that even the possible appearance of ℓ_2 inside a quotient of $C(\omega^\omega)$ was unknown [14].
- Argyros–Kanellopoulos again call the problem well known and open [2, Remark 4].
- Gasparis repeats the question in the source paper [6, p. 492].
- A 2020 MathOverflow formulation by Argyros still describes it as a well-known problem [10].
- The 2017 operator theorem gives the strong conclusions in theorems 3.1 and 3.4, but it proves superprojectivity rather than the subprojectivity required by theorem 4.1.

- The 2025 three-space paper develops new criteria for subprojectivity and superprojectivity but does not provide the missing quotient permanence [7].

This is a report of the search, not a claim that no inaccessible manuscript exists. The precise conclusion is that no published resolution was located in the supplied papers, arXiv searches, journal searches, bibliographic forward searches, or the recent subprojectivity literature.

5.2 The broad/isomorphic version has a counterexample

Under the convention $E^* \simeq \ell_1$, the answer is negative. Argyros–Motakis construct a hereditarily indecomposable $\mathcal{L}_\infty$ -space $\mathfrak{X}_{\text{nr}}$ containing no c_0 , no ℓ_1 , and no reflexive subspace [3]. Their Theorem 8.3 yields infinite-dimensional spaces

$$X_1 \twoheadrightarrow X_2 \twoheadrightarrow X_3$$

where all three are broad ℓ_1 -preduals with the scalar-plus-compact property, X_1 and X_3 are reflexively saturated, and the middle space is the no-reflexive-subspace construction. The preceding discussion identifies X_2 with $\mathfrak{X}_{\text{nr}}$ (or the corresponding version), and Corollary 3.12 proves that it contains no c_0 .

Hence $X_1 \twoheadrightarrow X_2$ is a quotient of a broad ℓ_1 -predual that is not c_0 -saturated. This fully settles the broad reading, but not the source question. Indeed, it cannot be converted naively into a strict example: if X_1 were a quotient of a strict ℓ_1 -predual, then the quotient map onto the c_0 -free infinite-dimensional space X_2 would be unconditionally converging and therefore compact by theorem 3.1, an impossibility.

6 Further structural reductions

The remainder records additional deductions that were not present in the supplied packet. They do not solve the problem, but they describe the shape of any strict counterexample.

6.1 A residual c_0 -splitting lemma

Lemma 6.1 (Transverse c_0 splitting). *Let X be a quotient of a real strict ℓ_1 -predual, let $Y \subset X$ contain no copy of c_0 , and let $q : X \rightarrow X/Y$ be the quotient map. If $A \subset X$ is isomorphic to c_0 , then:*

1. $q|_A$ is not compact;
2. there is a further $B \subset A$, $B \simeq c_0$, such that $q|_B$ is an isomorphism;
3. $q(B)$ is complemented in X/Y , and B is complemented in X .

Proof. Choose an isomorphism $U : c_0 \rightarrow A$ and put $a_n = Ue_n$. If $q|_A$ were compact, then $\|q(a_n)\| \rightarrow 0$, because (a_n) is weakly null. Passing to a subsequence, choose $y_n \in Y$ so that $\sum_n \|a_n - y_n\|$ is small enough for the principle of small perturbations. Then (y_n) is equivalent to a subsequence of the c_0 basis, contradicting the assumption on Y . This proves (1).

Since $A \simeq c_0$, every unconditionally converging operator from A is compact (apply theorem 3.1 to c_0 , or use property (V) and $c_0^* = \ell_1$). Thus the noncompact operator $q|_A$ is not unconditionally converging and fixes a copy $B \simeq c_0$, proving (2).

The quotient X/Y is separable, so Sobczyk’s theorem complements $q(B)$. If $P : X/Y \rightarrow q(B)$ is a projection and $S = (q|_B)^{-1}$, then SPq is a projection from X onto B . $\square$

Corollary 6.2 (Iterated peeling). *If a strict counterexample contains an infinite-dimensional c_0 -free subspace $Y \subset X$, then for every n there are complemented copies $B_1, \dots, B_n \simeq c_0$ and a complemented subspace $X_n \subset X$ such that*

$$X = B_1 \oplus \dots \oplus B_n \oplus X_n, \quad Y \subset X_n,$$

and X_n is again a quotient of a strict ℓ_1 -predual. Moreover, X_n/Y remains infinite-dimensional.

Proof. The identity on the infinite-dimensional X is noncompact, so theorem 3.1 gives a copy of c_0 in X . Apply theorem 6.1. The resulting projection onto B_1 has a kernel X_1 containing Y , and X_1 is a quotient of X via the complementary projection; hence it remains in the class of theorem 3.1. If X_1/Y were finite-dimensional, then Y would be complemented in X_1 , contradicting theorem 3.2. Iterate. $\square$

This produces arbitrarily long c_0 decompositions transverse to a bad subspace. It does not by itself produce a c_0 sequence in Y , because the norms of the successive projections need not be uniformly controlled. That loss of quantitative control is a concrete obstruction to turning the peeling argument into a proof.

6.2 A nonextendability phenomenon

Proposition 6.3 (Unconditionally converging extensions must fail). *Let $Y \subset X$ be infinite-dimensional and c_0 -free, where X is as in theorem 3.1. There exists a bounded, noncompact, unconditionally converging operator $T : Y \rightarrow c_0$. Every bounded extension $\tilde{T} : X \rightarrow c_0$ of T fixes a copy of c_0 ; in particular, no such extension is unconditionally converging.*

Proof. By the Josefson–Nissenzweig theorem, choose a normalized weak-star null sequence $(y_n^*) \subset Y^*$. Define

$$T(y) = (y_n^*(y))_{n=1}^\infty \in c_0.$$

Then T is bounded. It is not compact: $T^*e_n^* = y_n^*$ has norm one, whereas compactness would force this weak-star null sequence to have a norm-null subsequence. Since Y contains no c_0 , every operator defined on Y is unconditionally converging.

The separability of X and Sobczyk’s extension theorem provide a bounded extension $\tilde{T} : X \rightarrow c_0$. If $\tilde{T}$ were unconditionally converging, theorem 3.1 would make it compact, and then its restriction T would be compact, a contradiction. Hence every extension fixes c_0 . $\square$

This proposition isolates the hereditary gap. On the whole space, unconditionally converging means compact; on a hypothetical c_0 -free subspace, there are automatically noncompact unconditionally converging operators. Every extension is forced to acquire a c_0 copy outside the subspace.

6.3 The (WU) trichotomy

Odell calls a Banach space X a (WU) space if every normalized weakly null sequence in X has an unconditional subsequence. Every quotient of a space with a shrinking unconditional finite-dimensional decomposition has (WU) [11, Theorem A].

Theorem 6.4 (General (WU) trichotomy). *Let X have property (WU) . Every infinite-dimensional subspace $Y \subset X$ contains a copy of ℓ_1 , a copy of c_0 , or an infinite-dimensional reflexive subspace.*

Proof. Assume that Y contains neither ℓ_1 nor c_0 . If Y is reflexive, we are done. Otherwise, by Eberlein–Šmulian choose a bounded sequence with no weakly convergent subsequence. Rosenthal’s ℓ_1 theorem gives a weakly Cauchy subsequence (y_n) , which is not weakly convergent and hence not norm Cauchy. Select $n_1 < m_1 < n_2 < m_2 < \cdots$ and $\delta > 0$ with $\|y_{m_j} - y_{n_j}\| \geq \delta$, and set

$$z_j = \frac{y_{m_j} - y_{n_j}}{\|y_{m_j} - y_{n_j}\|}.$$

Then (z_j) is normalized and weakly null. Property (WU) gives an unconditional basic subsequence. Its closed span contains neither ℓ_1 nor c_0 . By James’s criteria, the unconditional basis is both shrinking and boundedly complete, so its span is reflexive [8, 13]. $\square$

Corollary 6.5. *If X has (WU) and contains no ℓ_1 , then*

$$X \text{ is } c_0\text{-saturated} \iff X \text{ contains no infinite-dimensional reflexive subspace.}$$

In particular, this applies to quotients of spaces with shrinking unconditional FDDs.

For a quotient X of countable $C(K)$, every subspace $Y \subset X$ has separable dual: X^* embeds into $C(K)^* = \ell_1(K)$ and the restriction map $X^* \rightarrow Y^*$ is onto. Hence no such Y contains ℓ_1 . For the first hard level $C(\omega^\omega)$, Odell also records Schlumprecht’s observation that a normalized weakly null sequence with an ℓ_1 spreading model generates a subspace containing c_0 [11, Section 3]. Therefore a c_0 -free, no-reflexive-subspace counterexample at that level would have to fail (WU) through a Maurey–Rosenthal type sequence, while simultaneously avoiding every ℓ_1 spreading model.

7 Why the natural closing arguments fail

The preceding results leave several tempting but invalid shortcuts.

1. **The Schur property does not pass to the dual of a subspace.** Although $X^* \subset \ell_1$ is Schur, for $Y \subset X$ the space Y^* is a quotient of X^* , and quotients of ℓ_1 can be arbitrary separable Banach spaces. Thus Y^* may be reflexive or James-like.
2. **Superprojectivity points in the wrong direction.** It controls infinite-codimensional superspaces and quotients. The desired conclusion is subprojectivity, which controls subspaces. Neither is a formal consequence of the other.
3. **Cantor–Bendixson induction has a nonclosed-range gap.** For countable K , the natural kernel $C_0(K \setminus K') \simeq c_0$ suggests transfinite induction. But the image of this kernel under an arbitrary quotient map need not be closed. The closure of a nonclosed operator range from c_0 can hide a c_0 -free separable space, so the exact sequence alone does not preserve saturation.
4. **The broad counterexample cannot simply be renormed.** The strict hypothesis supplies property (V) and hence $\mathcal{U} = \mathcal{K}$ after passing to quotients. The Argyros–Motakis quotient onto a c_0 -free space violates exactly this operator conclusion.
5. **Iterated splitting lacks uniform constants.** Theorem 6.2 gives arbitrarily many complemented c_0 pieces away from a bad Y , but the projection norms and isomorphism constants may diverge. Without uniform control, no bounded c_0 -sum projection or basis decomposition follows.

These are not merely technical annoyances: they identify the point at which each natural proof strategy requires genuinely new information.

8 Final status

1. The supplied complemented-subspace result is correct. Its only needed repair is the citation to Krulišová's Theorem 4.2.
2. A stronger known theorem gives $\mathcal{U}(X, Z) = \mathcal{K}(X, Z)$ for every quotient X of a strict real ℓ_1 -predual and every Z .
3. Under the broad convention $X^* \simeq \ell_1$, the literature contains a full counterexample, via Argyros–Motakis.
4. Under the strict convention intended by the source, and in particular for quotients of countable $C(K)$, no full proof or counterexample was located through 22 June 2026.
5. The strict problem is exactly the subprojectivity of these quotients. Any counterexample must support the transverse splitting and extension phenomena proved above, while avoiding the standard (WU) and spreading-model routes to c_0 .

References

- [1] D. E. Alspach, *A ℓ_1 -predual which is not isometric to a quotient of $C(\alpha)$* , Banach Spaces (Mérida, 1992), Contemp. Math. **144**, American Mathematical Society, 1993, 9–14.
- [2] S. A. Argyros and V. Kanellopoulos, *Determining c_0 in $C(K)$ spaces*, Fund. Math. **187** (2005), 61–93.
- [3] S. A. Argyros and P. Motakis, *The scalar-plus-compact property in spaces without reflexive subspaces*, arXiv:1608.01962, 2016.
- [4] J. Diestel, *Sequences and Series in Banach Spaces*, Graduate Texts in Mathematics 92, Springer, 1984.
- [5] E. M. Galego, M. González, and J. Pello, *On subprojectivity and superprojectivity of Banach spaces*, Results Math. **71** (2017), 1191–1205.
- [6] I. Gasparis, *New examples of c_0 -saturated Banach spaces*, Math. Ann. **344** (2009), 491–500.
- [7] M. González and J. Pello, *On the three-space property for subprojective and superprojective Banach spaces*, Mediterr. J. Math. **22** (2025), article 54.
- [8] R. C. James, *Bases and reflexivity of Banach spaces*, Ann. of Math. (2) **52** (1950), 518–527.
- [9] H. Krulišová, *Quantification of Pełczyński's property (V)*, arXiv:1509.06610, 2015.
- [10] S. A. Argyros, *Quotients of subspaces of $C(\alpha)$* , MathOverflow question 360068, 11 May 2020, <https://mathoverflow.net/questions/360068/quotients-of-subspaces-of-c-alpha>.
- [11] E. Odell, *On quotients of Banach spaces having shrinking unconditional bases*, Illinois J. Math. **36** (1992), 681–695.
- [12] A. Pełczyński, *Banach spaces in which every weakly unconditionally converging operator is weakly compact*, Bull. Acad. Polon. Sci. Sér. Sci. Math. Astronom. Phys. **10** (1962), 641–648.

- [13] H. P. Rosenthal, *A characterization of Banach spaces containing ℓ_1* , Proc. Natl. Acad. Sci. USA **71** (1974), 2411–2413.
- [14] H. P. Rosenthal, *The Banach spaces $C(K)$* , in Handbook of the Geometry of Banach Spaces, Vol. 2, North-Holland, 2003, 1547–1602.
- [15] A. Sobczyk, *Projection of the space (m) on its subspace (c_0)* , Bull. Amer. Math. Soc. **47** (1941), 938–947.